Data Analyst Assignment

OVERVIEW

This take-home assignment is intended to evaluate how you identify data quality issues, clean and prepare data, analyze operational trends, and communicate your process and findings.

Candidates whose submissions are selected to move forward will be invited to interview and present their work to executive leadership. Virtual interviews or presentations may be considered on a case-by-case basis.

DATASET

You have been provided with an Excel file containing Columbus Fire & EMS incident response data from 2023 through 2025. Each row represents a specific unit's response to an incident within the City of Columbus. The dataset includes incident details, unit response information, and various timestamps. Please refer to the Data Dictionary and other definitions at the end of this assignment to support you in your analysis.

INSTRUCTIONS

Please review, clean, and analyze the dataset using any platform or tool of your choice, including, but not limited to, Excel, Power BI, Tableau, SQL, Python, and/or R.

Based on your analysis, create a data visualization, dashboard, or visual summary that clearly communicates your key findings. You are not expected to analyze every possible issue or trend in the dataset. Focus on the areas where you feel the data supports the most meaningful insights.

Please also provide written documentation briefly explaining your data cleaning process, methodology, assumptions, data quality findings, conclusions, and/or recommendations.

REQUIREMENTS

1. Data Cleaning and Preparation

Assess the quality of the data and prepare it for analysis. This may include identifying missing, duplicate, inconsistent, incomplete, or unusual records.

Document your methodology along the way, including how you handled data quality issues, created calculated fields, excluded records from calculations, or made assumptions during the analysis.

2. Operational Analysis

Analyze the data to identify meaningful trends, patterns, and key measures related to incident response activity.

Examples of areas to consider include, but are not limited to:

- Incident volume over time, such as year, season, month, day of week, or hour of day
- Incident activity by incident type, station, unit, zip code, response priority, or other available fields
- Geographic patterns, such as high-volume zip codes, fire stations, or other location-based trends
- Response performance, such as alarm handling, turnout time, travel time, total response time, or other calculable measures
- Resource utilization, such as number of units dispatched per incident, multi-unit response patterns, unit or station workload, or differences in resource use by incident type

3. Findings and Recommendations

In addition to describing your methodology, please summarize your most important findings and recommendations. Recommendations should be practical, data-supported, and appropriate for a Fire & EMS executive leadership audience.

If a finding requires additional operational context, further validation, or additional study, please state that clearly.

DELIVERABLES

1. Analysis File, Dashboard, or Visual Summary

Submit the workbook, dashboard, code, report, or other file used to complete your analysis. Your submission should include at least one visualization, dashboard, or visual summary that clearly communicates your key findings.

If using code, SQL, or calculated measures, please include enough detail for reviewers to understand your major cleaning, transformation, and analysis steps.

2. Written Documentation

Submit notes or a brief report summarizing your methodology, assumptions, data quality findings, key findings, recommendations, and any limitations.

Documentation does not need to be lengthy, but it should be clear enough for reviewers to understand your conclusions and the steps taken to reach them.

DATA DICTIONARY

FIELD NAME	DESCRIPTION
Incident Number	Identifier assigned to the emergency incident or call for service
Incident Full Address	Full address or cross streets associated with the incident
Primary Station	Fire station primarily associated with the incident location
Cold Response	Indicates whether the response was non-emergency/nonpriority, meaning without lights and sirens
Incident Type Category	Broad classification of the incident type
Incident Type	Specific incident classification or call type
Unit Call Sign	Identifier for the responding unit (E=Fire Engine, L=Ladder Truck, M=Medic/Ambulance, B=Battalion Chief, S=Squad Truck)
PSAP DateTime	Date and time the call was received by the Public Safety Answering Point, or 911 call center
Alarm DateTime	Date and time the station or units were notified and dispatched to the incident
Enroute DateTime	Date and time the unit was marked enroute to the incident
Arrival DateTime	Date and time the unit arrived at the incident

OTHER DEFINITIONS

Alarm Handling: Call processing time, or the elapsed time between when the call was received and when the incident was dispatched for response

Turnout Time: The elapsed time between when the unit was dispatched and when the unit marked enroute to the incident

Travel Time: The elapsed time between when the unit marked enroute and when the unit arrived at the incident

Total Response Time: The elapsed time between when the call was received and when the unit arrived at the incident. This measure encompasses alarm handling time, turnout time, and travel time